

ABDUL RAFAY ZAHID

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PROFESSIONAL SUMMARY

Backend and systems engineer with ~3 years of experience across C++ networking, Python and TypeScript backends, data pipelines, and AI agent infrastructure. Scaled a C++ TLS WebSocket gateway from 200 to 5,000 concurrent clients, ported a ZTNA stack to Solaris and ARM, and built, as sole developer, an email infrastructure platform sending 200K+ emails/day across 1,000+ domains. Open sourced colloid-swarm, a multi-agent development scaffold adopted as the engineering harness for roughly 50 engineers. Works across engineering, sales, and client teams: shipped dashboards for a CEO and an SDR team, delivered client-specific builds for a banking deployment, and maintains tooling other engineers build on. Looking for backend and systems roles at companies building real technology, with a particular interest in robotics, ADAS, and computer vision.

EXPERIENCE

Softaims

May 2025 – Present

Software Engineer

- Designed and built, as sole developer, an enterprise email infrastructure platform sending **200K+ emails/day across 1,000+ domains**: a multi-channel campaign engine over KumomTA and Amazon SES edge nodes (A/B testing, threaded follow-ups, real-time delivery-state tracking), automated SPF/DMARC/DKIM lifecycle management, edge agent daemons across multi-region VPS nodes, and graduated warmup with Redis-locked cycles and blacklist monitoring. Response rates beat the Salesforce tool it replaced; domain onboarding dropped from hours to minutes.
- Built the platform's bi-directional Salesforce sync, an SDR productivity dashboard, and a CEO dashboard with data-freshness indicators, so sales and leadership work from the same delivery data as engineering.
- Built the voice onboarding module of Meridian, a service-business profitability platform: an ElevenLabs Conversational AI agent runs a spoken discovery interview while the backend owns conversation state through a slot-walk that picks the next unfilled value and the next question, keeping the agent to a single turn and making the whole interview script reproducible and testable offline.
- Rebuilt a food delivery and restaurant management platform serving **10,000+ daily orders across 200+ restaurants**: institutional co-payment and network discount workflows, pseudo-escrow payouts across vendors with milestone-based pricing, recurring and one-time billing across license tiers, real-time delivery tracking with location triangulation, and GenAI allergen and dietary tagging with recipe decomposition, which cut vendor support tickets by 35%. [app.bitley.com.au]
- Designed a RAG recommendation system in NextJS with OpenAI embeddings that matches student profiles to scholarships and colleges (**78% match satisfaction across 5,000+ students**), fed by Playwright scrapers collecting from 50+ sources at 10,000+ records/day and a NumPy/Pandas ETL pipeline in a FastAPI backend for validation across database migrations. [instudi.com]
- Architected an extensible deep research agent framework in LangChain with plug-and-play agentic workflows so other teams could stand up domain-specific research tools without rebuilding the plumbing; applied the same primitives to an LLM essay-feedback system with tool-based writing-coach agents and a prompt layer that updates at runtime from teacher guidance, rubrics, and sample references. [engagemo.ai]

Embryx Pvt Ltd (Invisily ZTNA)

Jun 2024 – Apr 2025

Associate Software Engineer

- Scaled a C++ gateway's TLS WebSocket capacity from **200 to 5,000 concurrent clients** through connection pooling, optimized network I/O, memory management, threading models, and bulk processing. Examples of changes made included custom non-blocking polling implementations, initializing arena memory upfront, and batching WebSocket lifecycles. Extreme-load tests exposed failure paths normal traffic did not, including log rotation temporarily removing a file between checks and reads and crashing a worker thread; fixed those paths to keep the gateway stable under sustained load.
- Took over a task that had been passed between engineers: porting the C++ ZTNA client to Solaris for a banking client, where the official tunnelling-API documentation was wrong. Reverse-engineered the real API behaviour and delivered the build that unblocked the deployment.
- Rewrote the core Python policy engine, the central decision-maker of the microservices architecture, from a functional to an object-oriented design, coordinating its interfaces with the Web, Desktop, Mobile, MFA, DB, and IoT components; re-architected policy evaluation from a cron job to an event-driven model, cutting redundant computation and database load by **29%** (cProfile).
- Implemented a Kafka streaming pipeline feeding packet-level access telemetry to ML models for both training and live inference, assigning dynamic risk scores to services, users, and devices across tenants; packaged the device profiler as a dependency-free Ubuntu Server binary via Cython.
- Ported a ZTNA gateway component from x86-64 Ubuntu Server to ARM (Raspberry Pi) for a client's self-contained on-premises deployment, and engineered the auto-update migration system that carried the gateway across the move to a multi-tenancy architecture.
- Rewrote MySQL connection pooling with RAII patterns via Python's ContextManager so every acquire is bound to a release, eliminating pool exhaustion in complex and exception-heavy flows; removed **2,000+ LoC** of duplication and cut code smells by **37%** (SonarQube).

Active Takeoff

Nov 2023 – Apr 2024

Software Engineer

- Wrote custom OCR software to parse tabular data from images across hundreds of documents into MySQL for enriched modeling, and indexed customer data in Elasticsearch for sub-millisecond search with image-based retrieval, wired into the existing frontend.

Cloud and App Development Intern

- Built PostgreSQL-backed APIs and Azure AD authentication for an e-commerce platform, then containerized its cloud deployment with Docker and Kubernetes.
- Contributed to a .NET government-portal backend and designed the GitHub Actions deployment path into the client's in-house data center.

PROJECTS

Colloid Swarm | *Multi-agent development scaffold*

github.com/Sarb0Z/colloid-swarm

- Portable multi-agent development scaffold, adopted as the company-wide engineering harness for roughly 50 engineers. Seven delegating personas route work across three cost and capability tiers, keeping planning and architecture at the top tier and running hostile review as an independent gate rather than self-assessment by the implementer.
- One engine-neutral core (instructions, skills, hook policies, MCP registry) with thin adapters for Claude Code, Codex, Kimi, and GitHub Copilot; an exporter transplants the scaffold into a target repository at a pinned commit.

Facial Emotion Recognition | *PyTorch, OpenCV, ONNX, Unity*

github.com/Sarb0Z/EmotionRecognitionModel

- Trained a convolutional neural network from scratch on the FER2013 training set (28,709 images, 7 emotion classes) and built a real-time inference loop in OpenCV: Haar-cascade face detection, 48x48 grayscale crop, per-frame prediction, and an on-frame emotion label. Exported the model to ONNX for on-device inference inside a Unity game.

NPI Discovery Service | *NestJS, Next.js, Redis, Terraform*

github.com/Sarb0Z/npi-discovery-service

- NestJS and Next.js monorepo for searching and bulk-exporting US healthcare providers from the public NPPES registry. A partition-aware collector splits by provider type, then narrows by postal prefix until each query fits upstream result limits, reporting incomplete partitions instead of silently truncating.
- Redis caching, a WebSocket gateway streaming bulk-job progress, correlation-ID tracing, and Terraform deploys.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, Bash**Frameworks & Libraries:** FastAPI, NestJS, Next.js, LangChain, LangGraph, Prisma, NumPy, Pandas, OpenCV, PyTorch, Cython**Data & Infra:** Kafka, BullMQ, Docker, Kubernetes, Linux, Terraform, AWS (SES, SNS, S3), GCP (Cloud Run, Cloud Functions, Firestore), Postgres, MySQL, MongoDB, Supabase, Firebase, Elasticsearch, Redis**Tools:** Git, GitHub Actions, MCP, ElevenLabs, Stripe, Playwright, Postman, SonarQube, cProfileEDUCATION

FAST NUCES

Aug 2020 – Jun 2024

Bachelor of Science in Computer Science

- Coursework: Artificial Intelligence, Computer Vision, Advanced Database Concepts, Parallel and Distributed Computing.
- Final Year Project: real-time OpenCV facial-expression feedback pipeline on a CNN trained from scratch on FER2013.

Karachi Grammar School

Aug 2017 – May 2019

A-Levels: Physics, Maths, Computer Science

- Founder and President, World Affairs Society: established the society and launched its flagship event, Renaissance 19 (planning, speakers, logistics).